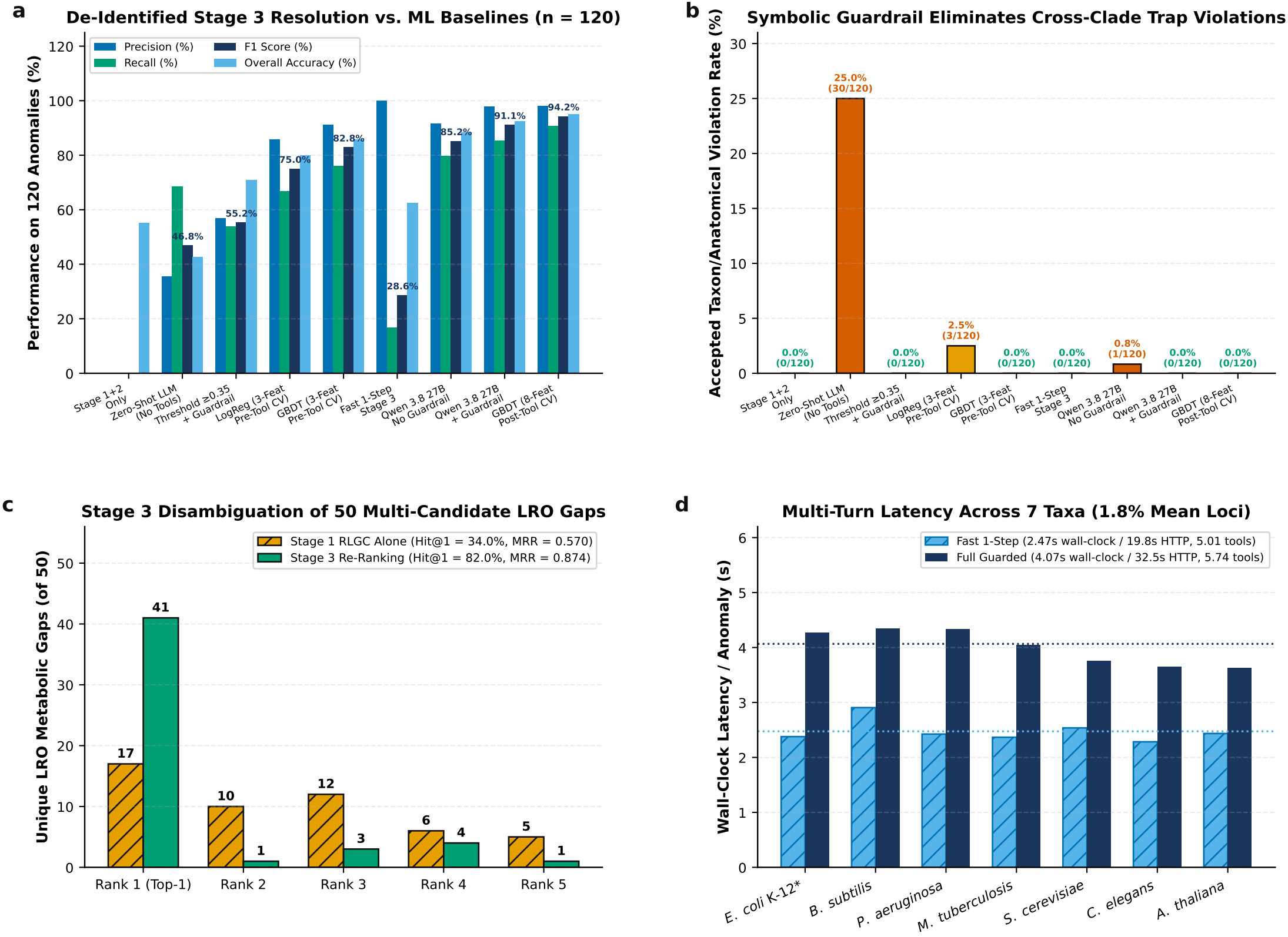

a
De-Identified Stage 3 Resolution vs. ML Baselines (n = 120)
Performance on 120 Anomalies (%)
Precision (%)
Recall (%)
F1 Score (%)
Overall Accuracy (%)
46.8%
55.2%
75.0%
82.8%
28.6%
85.2%
91.1%
94.2%
Stage 1+2 Only
Zero-Shot LLM (No Tools)
Threshold ≥0.35 + Guardrail
LogReg (3-Feat Pre-Tool CV)
GBDT (3-Feat Pre-Tool CV)
Fast 1-Step Stage 3
Qwen 3.8 27B No Guardrail
Qwen 3.8 27B + Guardrail
GBDT (8-Feat Post-Tool CV)
b
Symbolic Guardrail Eliminates Cross-Clade Trap Violations
Accepted Taxon/Anatomical Violation Rate (%)
25.0% (30/120)
0.0% (0/120)
2.5% (3/120)
0.8% (1/120)
c
Stage 3 Disambiguation of 50 Multi-Candidate LRO Gaps
Unique LRO Metabolic Gaps (of 50)
Stage 1 RLGC Alone (Hit@1 = 34.0%, MRR = 0.570)
Stage 3 Re-Ranking (Hit@1 = 82.0%, MRR = 0.874)
17
41
10
1
12
3
6
4
5
1
Rank 1 (Top-1)
Rank 2
Rank 3
Rank 4
Rank 5
d
Multi-Turn Latency Across 7 Taxa (1.8% Mean Loci)
Wall-Clock Latency / Anomaly (s)
Fast 1-Step (2.47s wall-clock / 19.8s HTTP, 5.01 tools)
Full Guarded (4.07s wall-clock / 32.5s HTTP, 5.74 tools)
E. coli K-12*
B. subtilis
P. aeruginosa
M. tuberculosis
S. cerevisiae
C. elegans
A. thaliana